

Progress Report

Day 21 — Kanban Board Implementation

Task Management MERN Stack Platform

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Tasks Completed

Per-Team Kanban Board

- Implemented a Kanban board inside the Team Detail page.
- Tasks are grouped into three workflow columns:
 - To Do
 - In Progress
 - Done
- Tasks are loaded dynamically based on the selected team.

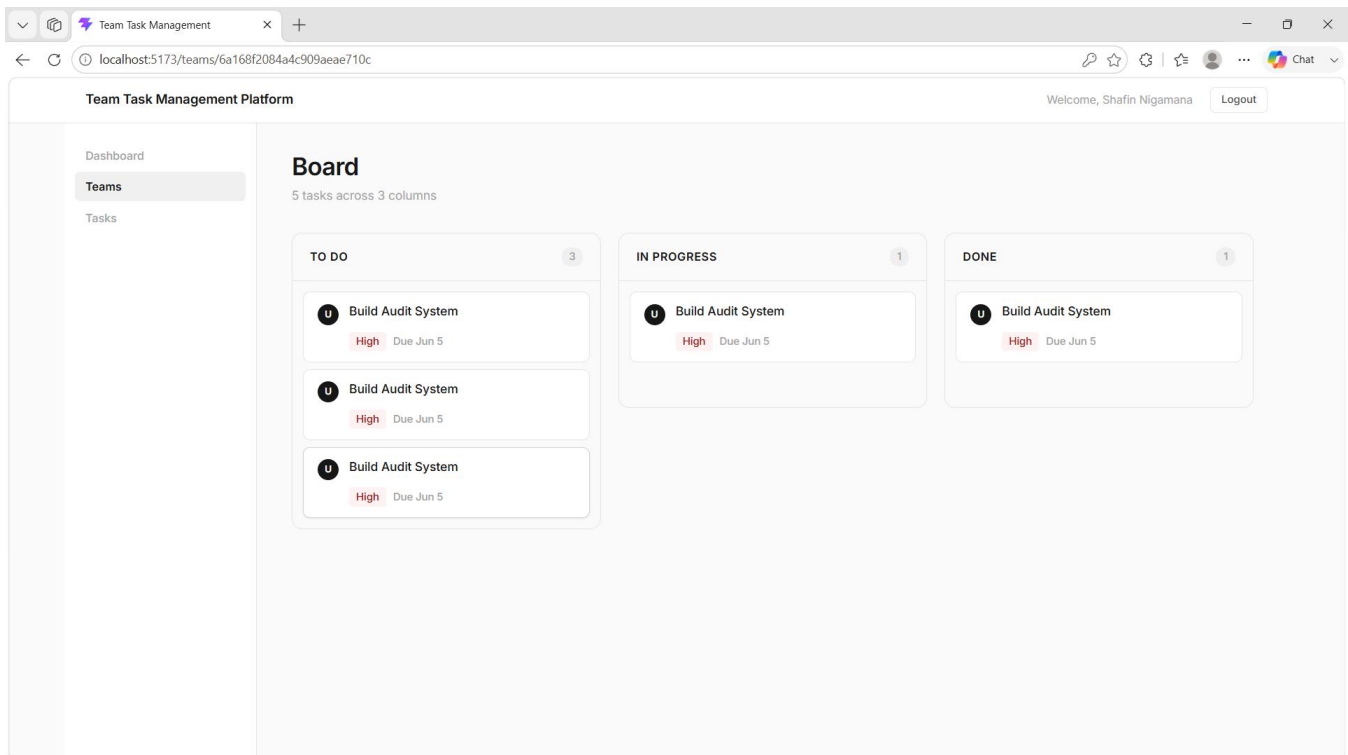


Figure 2.1 — Per-team Kanban board displaying tasks grouped by workflow status.

Drag-and-Drop Integration

- Integrated drag-and-drop functionality using `@dnd-kit`.
- Configured draggable task cards and sortable workflow columns.
- Added status update handling when tasks are moved between columns.

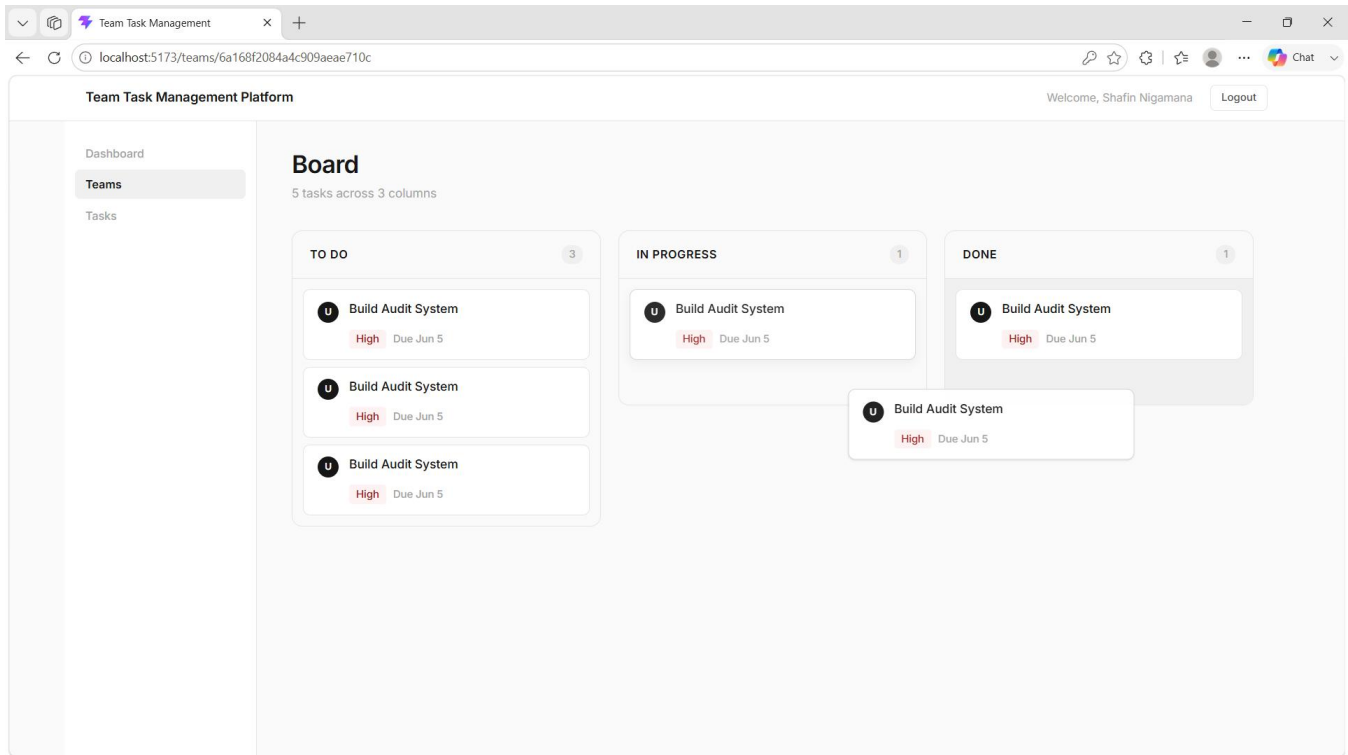
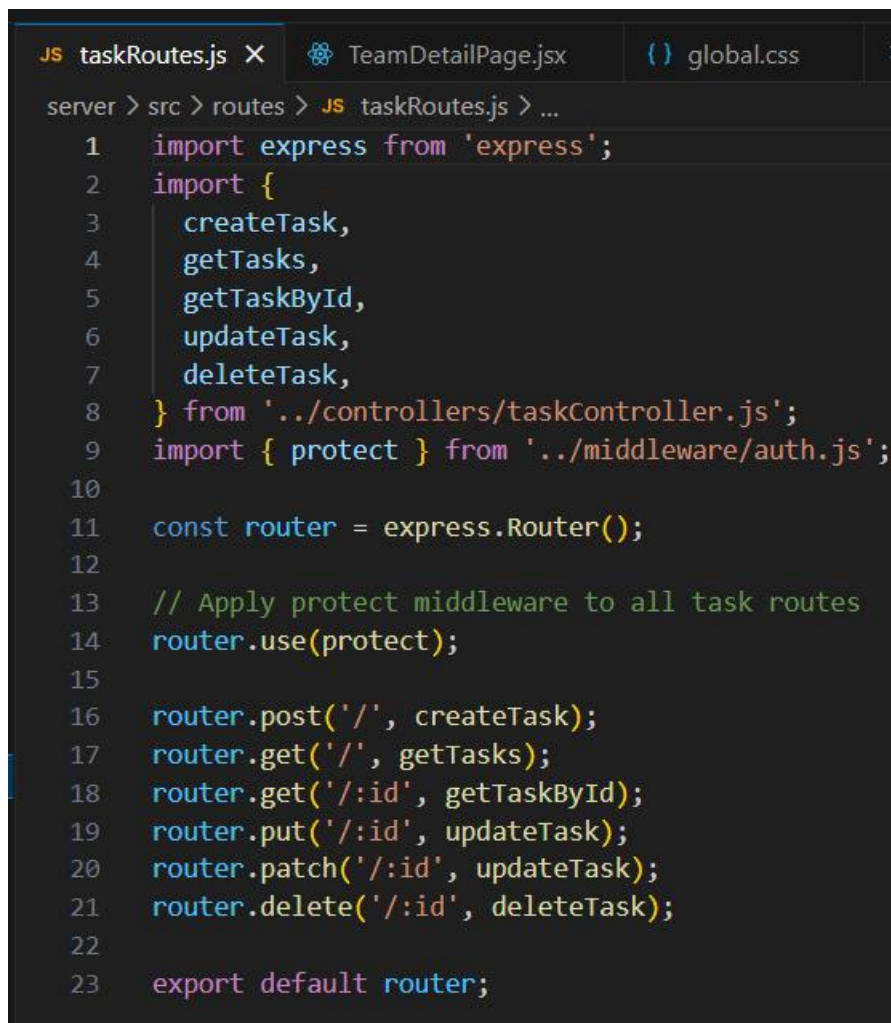


Figure 2.2 — Drag-and-drop functionality implemented for task workflow management.

Task Status Update API

- Added support for task status updates through the task API.
- Implemented PATCH endpoint integration for workflow transitions.
- Connected frontend task movement actions with backend status updates.



```
JS taskRoutes.js X TeamDetailPage.jsx {} global.css
server > src > routes > JS taskRoutes.js > ...
 1 import express from 'express';
 2 import {
 3   createTask,
 4   getTasks,
 5   getTaskById,
 6   updateTask,
 7   deleteTask,
 8 } from '../controllers/taskController.js';
 9 import { protect } from '../middleware/auth.js';
10
11 const router = express.Router();
12
13 // Apply protect middleware to all task routes
14 router.use(protect);
15
16 router.post('/', createTask);
17 router.get('/', getTasks);
18 router.get('/:id', getTaskById);
19 router.put('/:id', updateTask);
20 router.patch('/:id', updateTask);
21 router.delete('/:id', deleteTask);
22
23 export default router;
```

Figure 2.3 — Task status update request integrated with Kanban board interactions.

Task Card Enhancements

- Added task cards displaying:
 - Task title
 - Priority badge
 - Due-date badge
 - User avatar placeholder
- Applied consistent styling aligned with the existing design system.

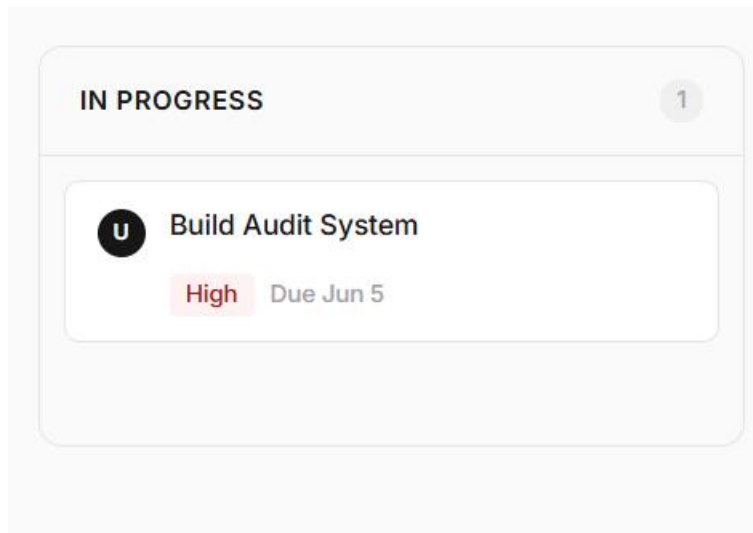


Figure 2.4 — Task card displaying assignment indicator, priority level, and due-date information.

Real-Time Data Refresh

- Implemented automatic task refresh using a 5-second polling interval.
- Added cleanup handling to prevent unnecessary background requests.
- Ensured task status changes remain synchronized with backend data.

```
// Initial fetch + 5-second polling
useEffect(() => {
  setLoading(true);
  setError(null);
  hasLoadedOnce.current = false;
  fetchTasks();

  const interval = setInterval(fetchTasks, 5000);
  return () => clearInterval(interval);
}, [fetchTasks]);

// — Drag and drop —
```

Figure 2.5 — Polling mechanism implemented for periodic task synchronization.